

Pros and Cons

Advantages of Naive Bayes

1. This algorithm works very fast and can easily predict the class of a test dataset.
2. You can use it to solve multi-class prediction problems as it's quite useful with them.
3. Naive Bayes classifier performs better than other models with less training data if the assumption of independence of features holds.

Disadvantages of Naive Bayes

1. If your test data set has a categorical variable of a category that wasn't present in the training data set, the Naive Bayes model will assign it zero probability and won't be able to make any predictions in this regard.
2. It assumes that all the features are independent. While it might sound great in theory, in real life, you'll hardly find a set of independent features.